

Update – Sep 18

JC Vaught

Fellowships

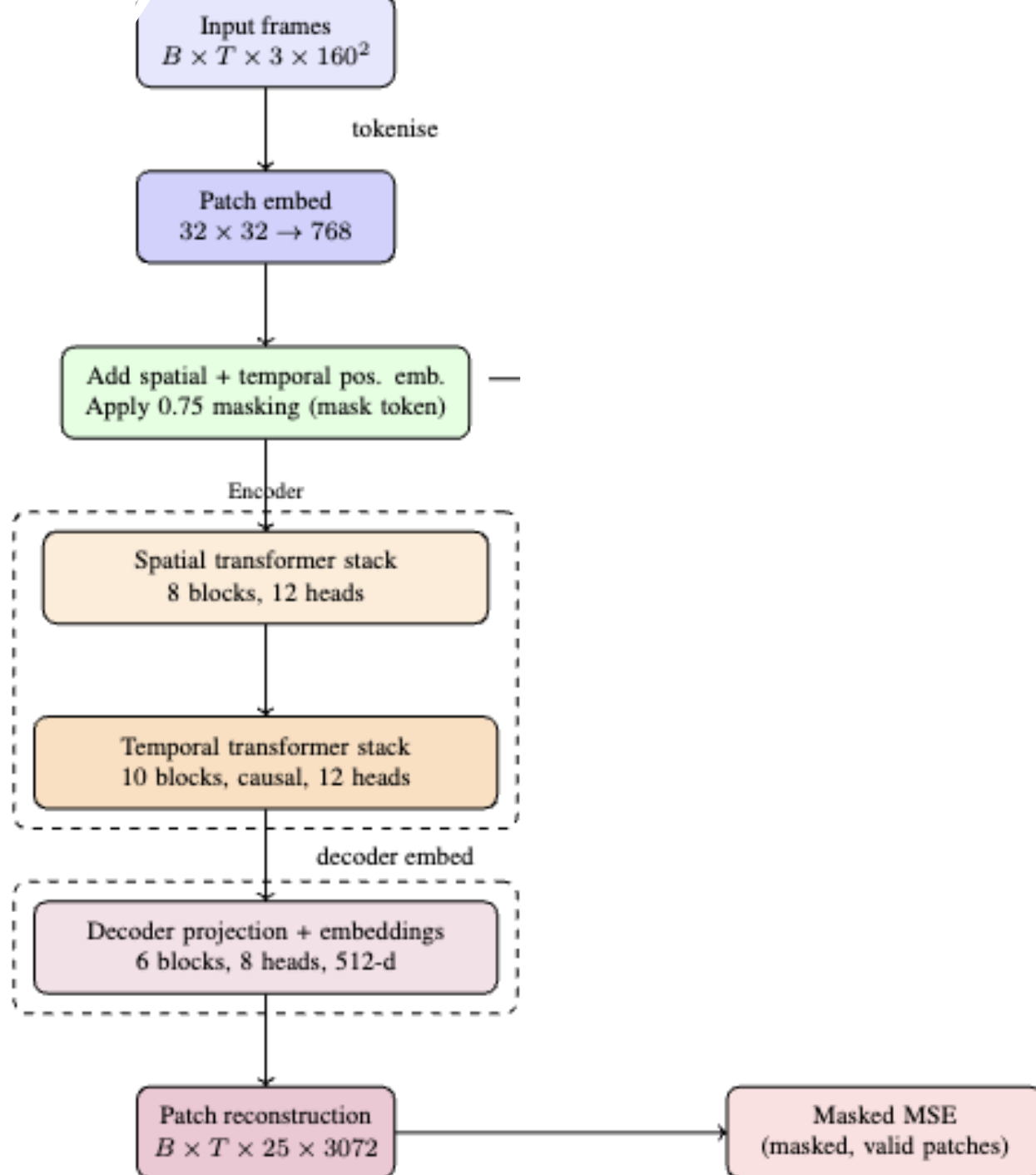
- DOD SMART – Dec 5
 - 1 in 12 approval; avg GPA 3.7; up to \$46k/yr stipend; 5yr funding
 - <https://www.smartscholarship.org/apply>
- NDSEG(National Defense) – Nov 15
 - 1 in 50 approval; \$40.8k stipend; 3 yr funding; Merit-based
 - <https://applyndseg.sysplus.com/Login>
- NSF GRFP – Nov 13
 - 1 in 5 approval; \$37k stipend; 3 yr funding; Merit-based
 - <https://www.research.gov/grfp/Login.do>
- DOE CGSF – Jan 16??? Not available
 - 1 in 20 approval; \$45k stipend; 4yr* funding; Merit-based
- Hertz fellowship – Oct 31
 - 1 in 40 approval; ~\$35k stipend; 5yr funding; Merit-based(but not really)
 - <https://fellowship.hertzfoundation.org/apply/>
- AIAA – Jan 31
 - ~1 in 33 odds
 - \$1k to \$10k
- ASME – Feb 26
 - ~1 in 10 odds
 - \$0.5k to \$10k

Progress

- Last week
 - Attempted Temporal-MAE for gain prediction
 - Full Frame masking
- This week
 - Rewrote to add variable context design
 - Adds more frames as they become available(i.e. real time)
- Now/Next week
 - Adding Full-Resolution capabilities
 - Sharding model across GPUs(running into OOM issues)

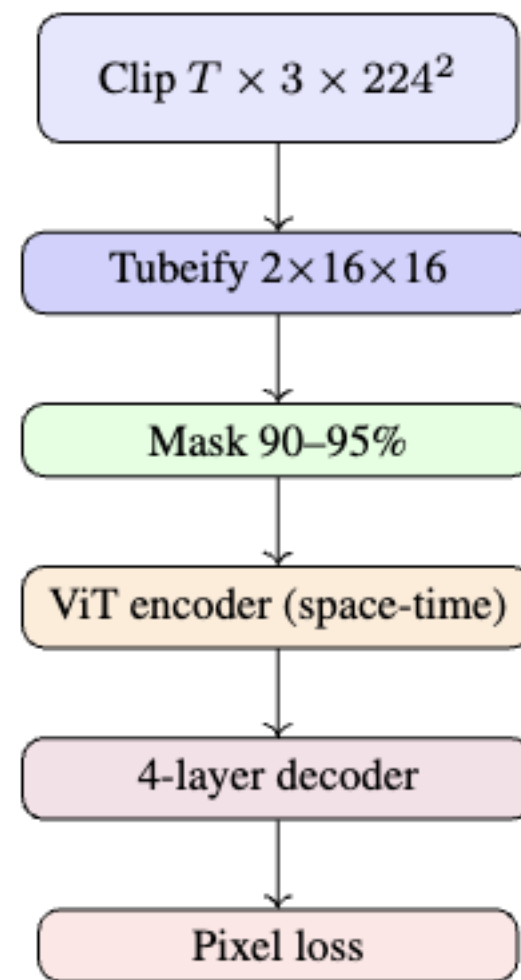
Architecture

- Similar to



Architecture VideoMAEv1

- What my system is based on basically

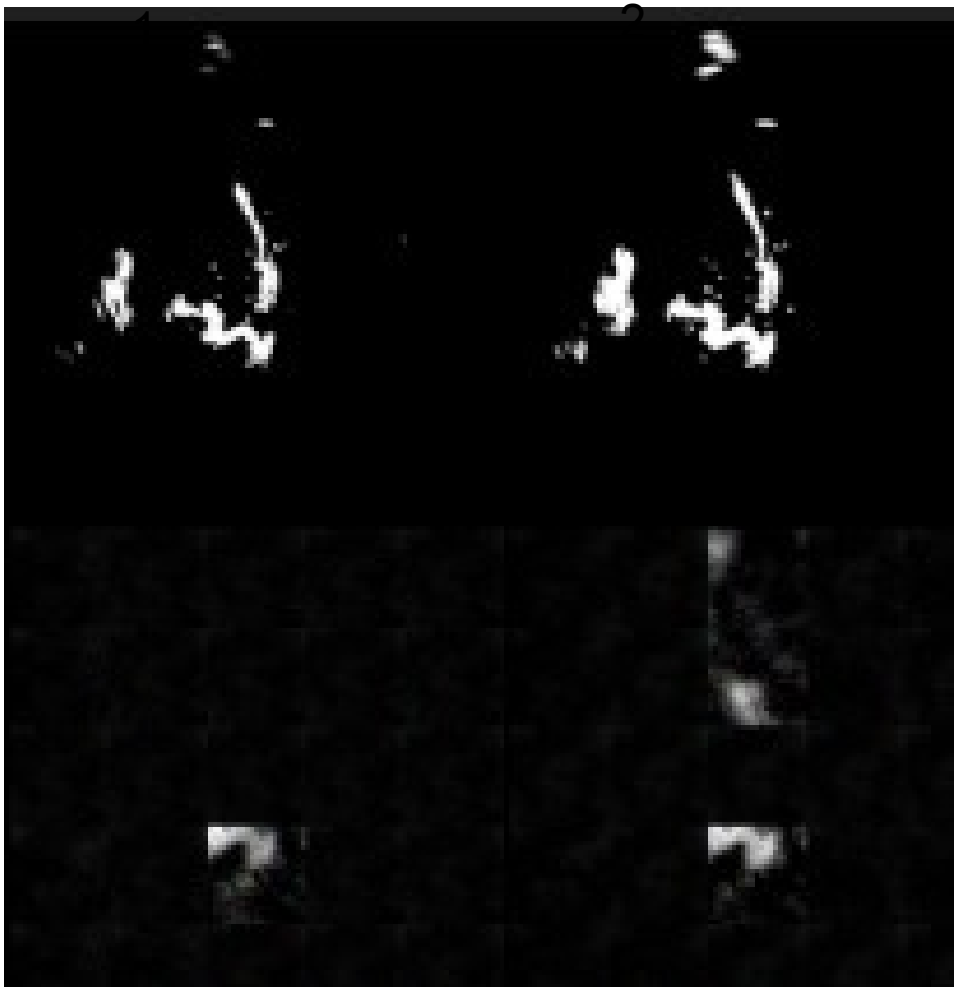


(a) VideoMAE[9]

Variable Context

Frame

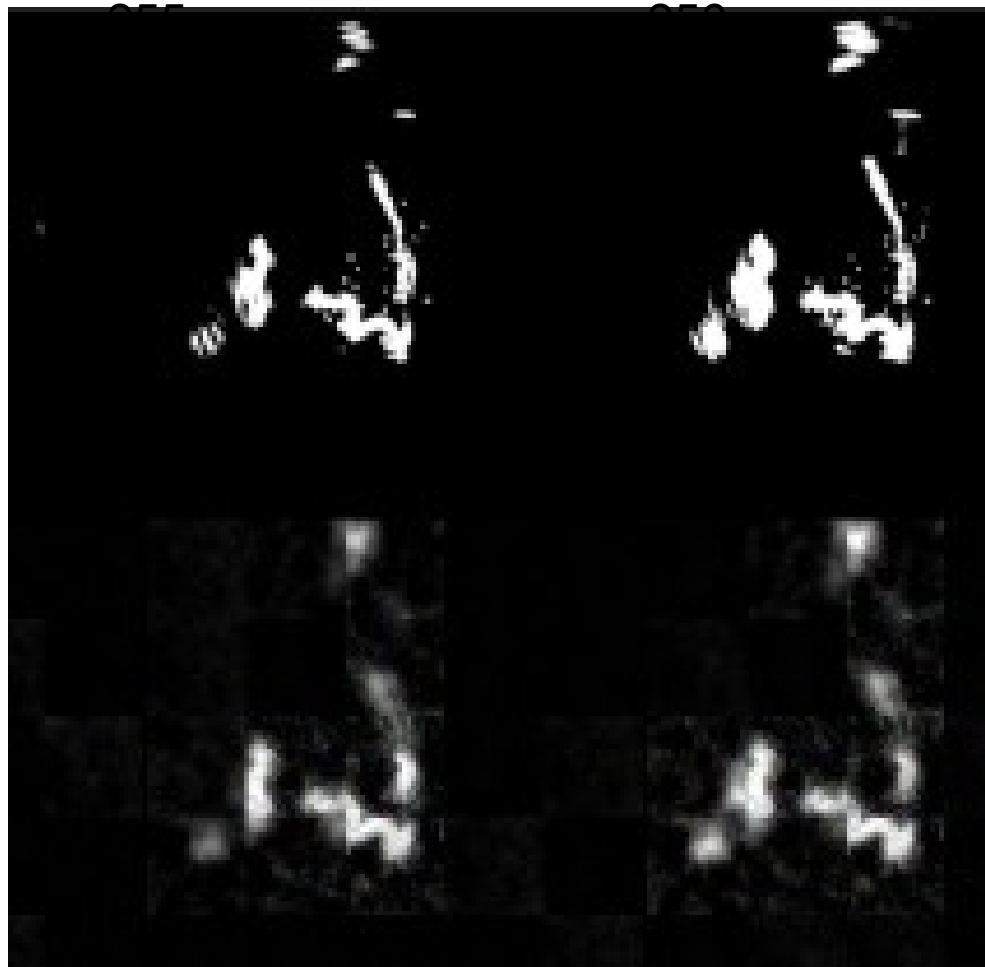
Frame



Predicted output

Frame

Frame



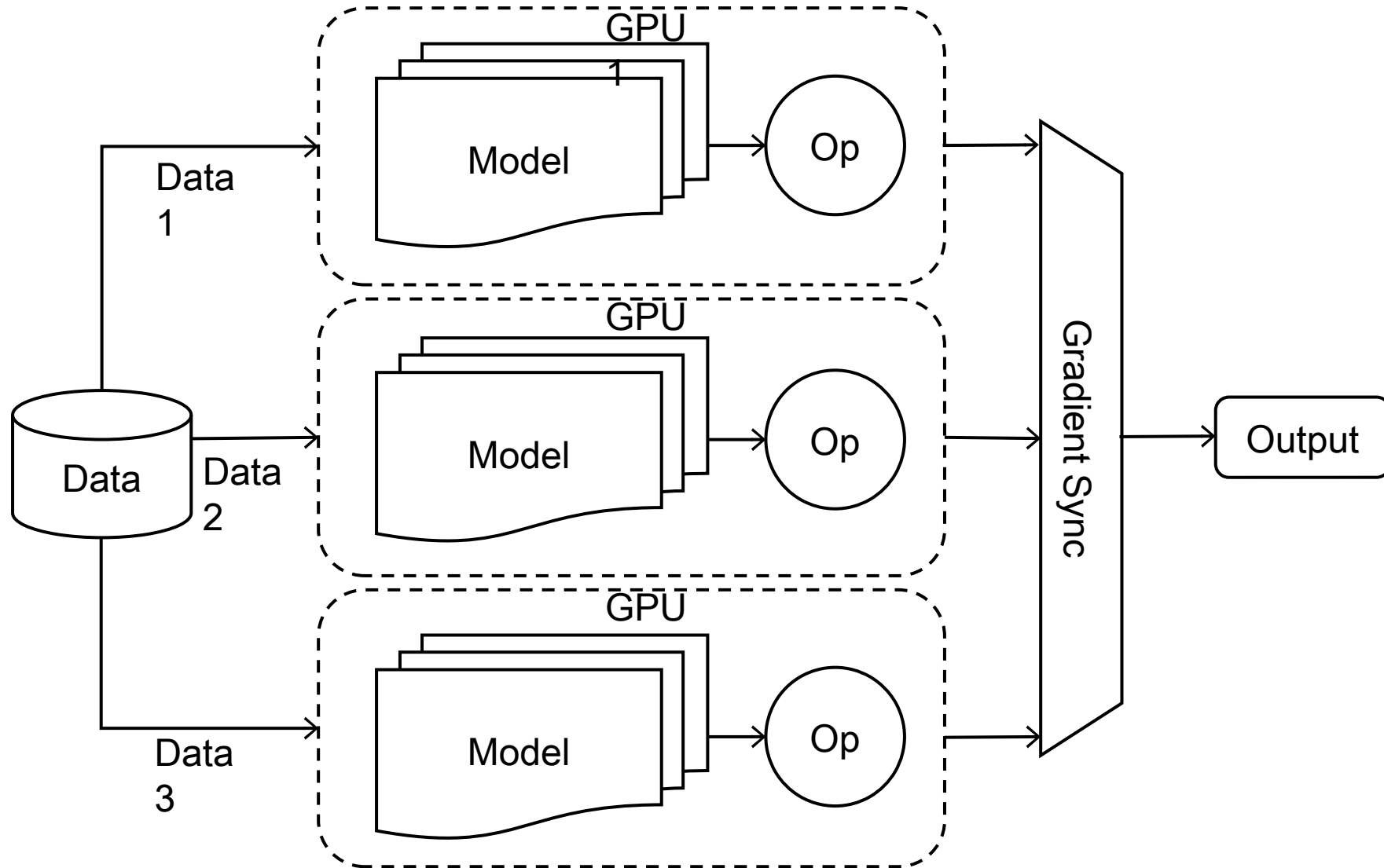
Predicted output

Splitting
Load up
between
GPUs

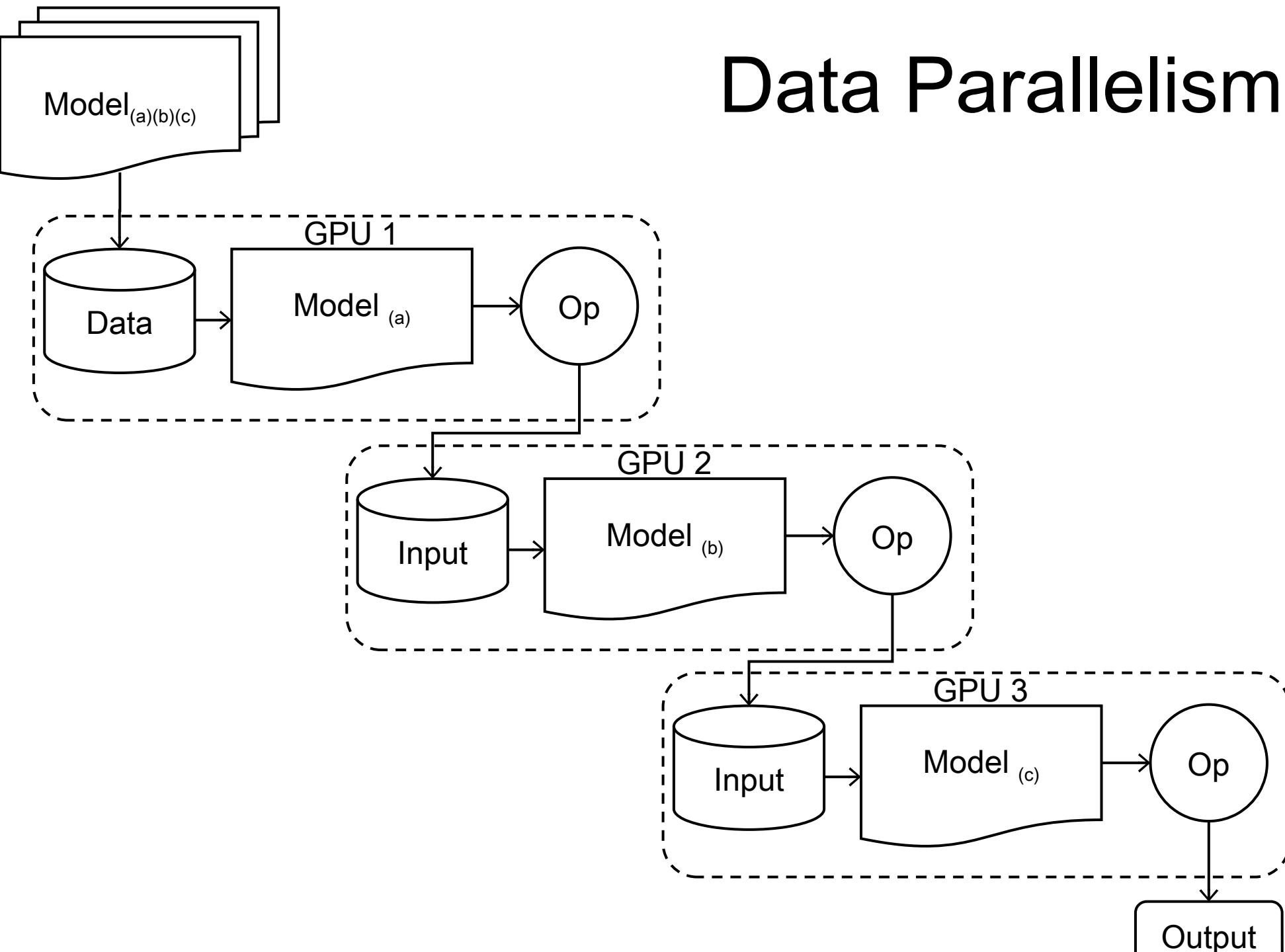
• **4 Options**

- Data Parallelism
- Model Parallelism
- Pipeline Parallelism
- Tensor Parallelism

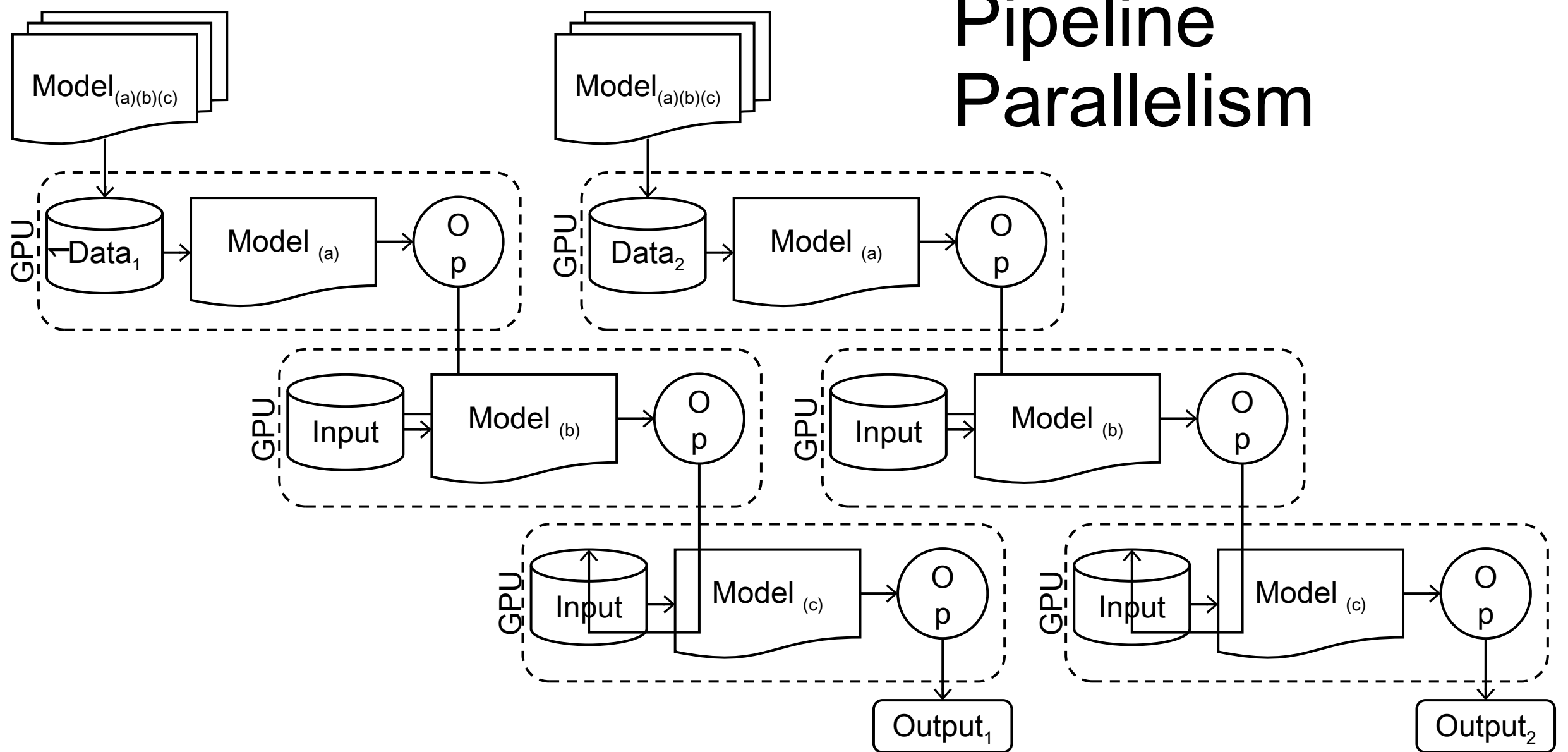
Data Parallelism



Data Parallelism



Pipeline Parallelism



Time
e

→

Tensor Parallelism

